DAURES GREEN HYDROGEN VILLAGE

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# Potentials of P2X in Namibia – Biogenic Carbon from Encroacher Bush for Clean Fuel Production

## NEGATIVE EMISSIONS THROUGH BIOGENIC CARBON

Climate protection and climate emission reduction are the challenges of today in a modern society and worldwide. Biogenic carbon from biomass may contribute to climate protection in various ways (Fig. 1). Through biogenic carbon even negative emissions can be generated. These processes are known as "Bioenergy carbon capture and storage (BECCS)".

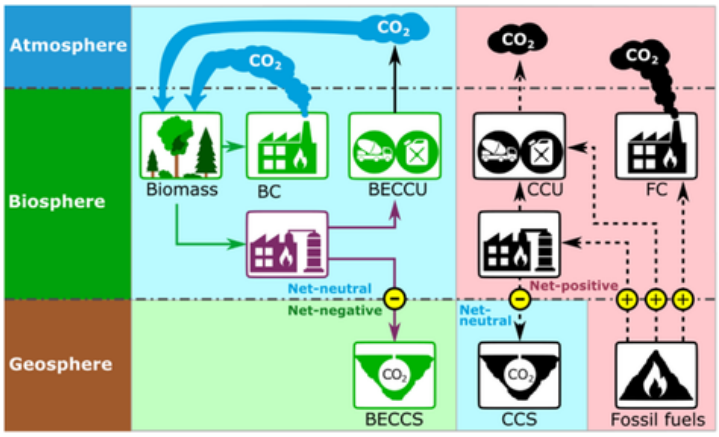


Figure. 1: Schematic diagram of carbon dioxide cycles in different bioenergy utilization routes (source: IEA bioenergy, June 2020)

For this effect the biogenic carbon needs to be separated and stored permanently in a suitable form and site. The production and storage of bio-char is a commonly used process leading to negative emissions.

Through processes called "Bioenergy Carbon Capture and Use" (BECCU) a circulation and more efficient use of biogenic carbon is achieved, but eventually the carbon is released into the environment again. A typical use as BECCU is the production and use of clean fuels e.g. through synthesis.

## ENCROACHER BUSH AND THE OTJIKOTO BIOMASS POWER STATION (OBPS)

Namibia has large natural resources, especially high solar and wind potentials. Also, biomass is available in large quantities e.g. in the bush encroachment. It represents a valuable resource, but particularly it is also a valuable stock of biogenic (clean) carbon. Over 45 Mio. hectares of land in Namibia are covered with various woody bush species. The bush encroachment is party so dense, that theintegrity and diversity of the naturalhabitat is in danger. Consequently, and over many years, a large effort has been made to reduce the bush biomass to a tolerable level and use it for various purposes e.g. charcoal, fodder or bioenergy [1].

The planned 40 MW Otjikoto Biomass Power Station (OBPS) from NamPower[1] would utilize encroacher bush within a 100 km operating radius.

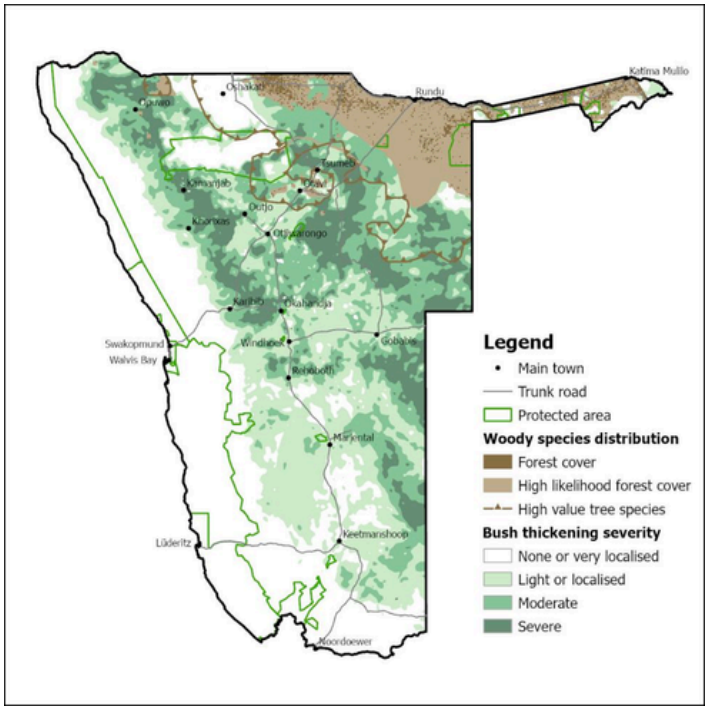


Figure 2: Bush encroached areas (acc. to van der Waal, 2024 [3]). The harvesting area around the Otjikoto Biomass Power Station (OBPS) would be around 100 km.

Figure. 3: Encroacher bush landscape in Namibia (photo: LE)

In the context of current climate and environmental challenges the biogenic (clean) carbon can also be used to produce clean fuels. In combination with green hydrogen to be produced in Namibia this is a promising option. Depending on the end use, the biogenic carbon can also contribute to "negative" greenhouse gas emissions.

## CLEAN CARBON AVAILABILITY IN NAMIBIA

The amount of carbon available at the OBPS plant is dependent on the species composition of the delivered biomass fuel. The specific carbon content of the main bush species is shown in Table 1.

| Species | C-Content (% DM) |
|---|---|
| Terminalia sericea | 40,2 – 50,7 |
| Acacia mellifera | 473 |
| Dichrostachys cinerea | 493 |
| Colophospermum mopane | 51,9 – 58,9 |

Table 1: Carbon content of typical species of the encroacher bush biomass.

Using the total dry weight values of standing wood within a 100 km operating zone, the bush biomass is estimated at 15,31 million tons, equivalent to approximately 7.8 million tons of carbon. Under full oxidation, this represents a maximum potential of around 28.7 million tCO2.

## CLEAN FUEL PRODUCTION IN NAMIBIA

Fuels for combustion engines in cars, trucks, ships or planes consist out of Hydrogen and Carbon (hydrocarbons). The distinction between petrol, diesel and kerosene is mainly due to the chain-length of the carbon skeleton (Table 2)

| Fuel type | C-chain length | properties |
|---|---|---|
| light end | C1-C8 | very volatile, highly flammable gases |
| petrol | C7-C9 | light, volatile, low boiling point |
| jet fuel | C9-C16 | medium volatility, temperature stable |
| diesel | C12 - C18 | heavy, high boiling points, less volatile |
| heavy cut | C16+ | heavy oil, lubricants, paraffines |

Table 2: Chain length and properties of typical fuels and hydrocarbons

For the synthetic production of clean fuels a clean carbon and a clean hydrogen source is necessary. Namibia is particularly suited to provide both elements in clean forms in abundancy. Clean hydrogen can com from the Green Hydrogen Projects at the coast, and clean carbon from the encroacher bush. However, they are not located well enough in the same area.

## P2X PRODUCTION FROM CLEAN MOLECULES

Clean Hydrogen or "Green Hydrogen" is produced via electrolysis from renewable energy sources (i.e. wind and solar). A number of exciting projects are developed in Namibia at present. Clean Carbon is available in large amounts in the Encroacher Bush in Namibia. Clean fuel production is an attractive option to use both elements. However, strict standards and certification schemes need to be met to comply with market regulations.

In principle various pathways to generate negative emissions or to reduce emissions are possible. Figure 4 shows the principle pathways to use biogenic carbon for climate protection and emission reduction.

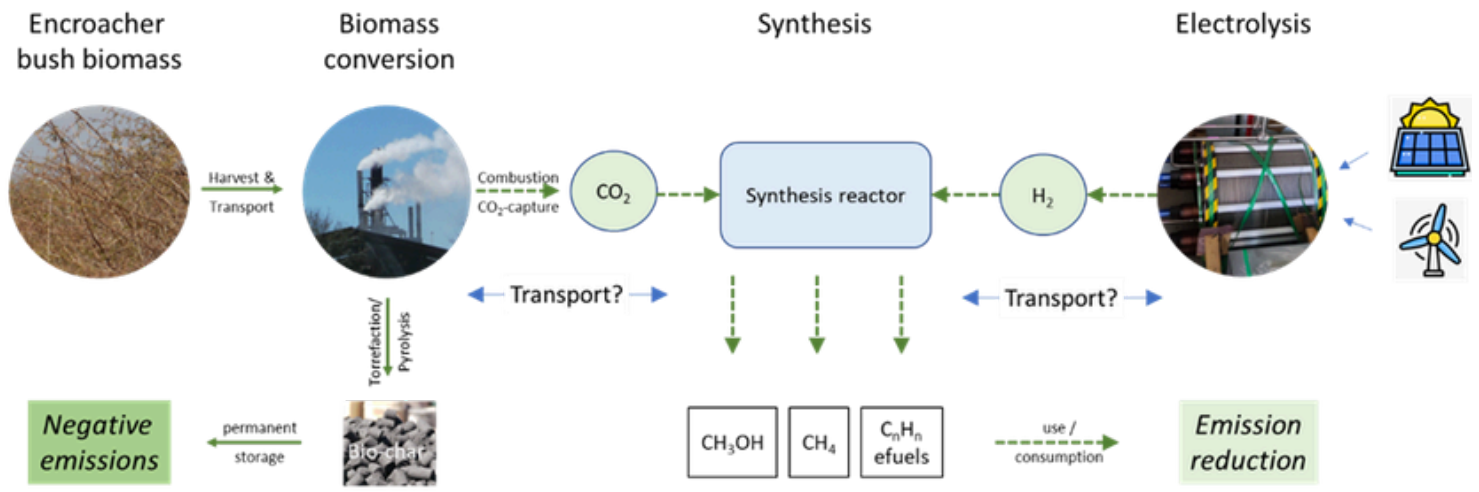


Figure. 4: Principle pathways to generate negative emissions from biogenic carbon i.e. encroacher bush or to contribute to emission reduction in the fuel sector.

The use of biogenic carbon in pathways such as electricity generation, methanol, SNG (substitute natural gas) or efuel production (BECCUS) contributes to a higher carbon efficiency through carbon circularity and to reduced greenhouse gas emissions of this process. But it does not qualify for 'negative' emissions.

## FISCHER-TROPSCH (FT) SYNTHESIS

The production of clean fuel is carried out in a reactor vessel. The most well known reaction is the Fischer-Tropsch Reaction developed around 100 years ago in 1925 by Franz Fischer and Hans Tropsch. In this reaction carbon (mainly as CO) and hydrogen are brought together at certain ratios, and at defined temperatures and pressures (Table 3) and with specially adapted catalysts.

| Parameter | Unit |
|---|---|
| H2/CO | 2:1 - |
| Temperature | 200 °C |
| Druck | 25 bar |

Table 3: Main reaction parameters of a Fischer-Tropsch reaction

The result of the FT-reaction is a mixture of components with different C-chain length. This "Syncrude" is then separated in a distillation process into the desired products inclusive SAF (sustainable aviation fuels).

## CONCLUSION

To produce clean fuels a sustainable hydrogen and a clean carbon molecule are the basic prerequisites. And Namibia has both! The high renewable potential can be used for green hydrogen production and several projects are being developed (DAURES, Cleanergy, Hyphen, etc.). And the encroacher bush are a vast resource of clean carbon.

The challenge is to bring both together and develop suitable production pathways and economic business models for harvesting and clean production at a large scale. A process for using biogenic carbon from encroacher bush biomass for P2X-processes is presented in GIZ 2022[2].

## REFERENCES

[1]An overview of current uses and importantstakeholders is providedat www.n-big.org/
[2] Development of a sustainable carbon carrier for PTX use. From Namibia to a global market. GIZ Oct. 2023.
[3] van der Waal, C.: 2024: Review and Consolidation of Namibia's Bush Thickening Map: Final Report on Bush
[4] Maps Project. Windhoek, Namibia. URL: https://the-eis.com/elibrary/search/30969.

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https://github.com/IER-Hy4Daures/Fact-Sheets

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